1 **(a)** $g(x) = f(x) + 7$ **[M1]**

$g(x) = 2x + 2$ **[A1]**

(b) Translation **[B1]**

by vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ **[B1]**

[Total Marks: 4]

2 **(a)** $g(x) = f(x + 2) = 3(x + 2)^2$ **[M1]**

$3(x^2 + 4x + 4)$ **[M1]**

$3x^2 + 12x + 12$ **[A1]**

(b) $(-2, -8)$ **[B2]**

(Allow B1 for each correct coordinate)

(c) Substitutes $x = 0$ into $h(x) = 3(x + 2)^2 - 8$ (or similar valid method) **[M1]**

4 **[A1]**

[Total Marks: 6]

3 **(a)** **(i)** $\frac{20}{x} - 4$ **[B1]**

(ii) $\frac{20}{x} - 4 = 0$ **[M1]**

$x = 5$ **[A1]**

(b) $h(x) = f(x - k) = \frac{20}{x - k}$ **[M1]**

$\frac{20}{8 - k} = 5$ **[M1]**

$k = 4$ **[A1]**

(c) Vertical asymptote: $x = 3$ **[B1]**

Horizontal asymptote: $y = 7$ **[B1]**

[Total Marks: 8]